

Dynamic Multi-UAV Path Planning for Multi-Target Search and Connectivity

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Abstract—In this work, we propose and analyze multi-drone path planners for multi-target search and connectivity. The goal of the unmanned aerial vehicle (UAV) mission is to search an unknown area to detect, connect and monitor multiple randomly distributed targets to the ground control station (GCS) while maintaining the connectivity of the UAVs to GCS. To this end, we propose to use two types of UAVs: search and relay. The search drones scan the area via onboard sensors, whereas relay UAVs provide connectivity. We propose three different responses to target detection with increasing adaptability: (i) follow pre-planned paths and inform GCS when possible, (ii) follow pre-planned paths and inject new UAVs to monitor the detected targets, (iii) assign a search UAV to monitor target, and re-plan remaining UAV paths. Furthermore, we implement multi-objective optimization-based planners for single-type UAVs, where the paths are optimized in terms of total coverage time and percentage connectivity. Our results show that less adaptive schemes detect all targets faster; however, the connectivity of the UAVs and the targets to GCS is significantly better for the adaptive schemes. The joint-optimization methods, on the other hand, in their basic form, trade-off detection time to improve connectivity. When they are used in conjunction with relays, the connectivity performance is improved; however, the overall mission times may be up to 40% higher than our proposed methods, depending on the number of UAVs and targets.

Index Terms—Drone networks, unmanned aerial vehicles, relay positioning, path planning, maintaining connectivity, multi-target.

I. INTRODUCTION

MULTI-DRONE systems can be very beneficial for search and rescue (SAR) operations during disasters, where many tasks are of interest, such as area surveillance, detection and monitoring of multiple targets and network provisioning [1]. Drones have become very versatile with their ability to be utilized as sensors, aerial base stations, or relays. In disaster scenarios, where relying on existing infrastructure might not be possible, fast data collection about the disaster area and data delivery to the rescue personnel are extremely important.

There might be various targets of interest, including damaged buildings, damaged roads, and victim clusters spread across the disaster area. Using a collaborative multi-drone system, large areas can be searched for targets, and UAVs can exchange valuable

information about detected targets, including their locations, status, and potential hazards, through connectivity maintenance. Consequently, the situation can be assessed more efficiently, and the targets may be responded to accordingly. Each of these tasks; i.e., target search, target connectivity to the backbone, UAV-UAV connectivity for team behaviour, UAV-to-ground control connectivity for data delivery or mission coordination, constitute important research problems. These problems are generally handled separately, where single type UAVs are utilized, which operate as aerial base stations, sensors and/or relays. In [2], [3], [4], [5], [6], [7], [8], [9], [10], [11], the main goal is network coverage provisioning to ground nodes (e.g., sensors, mobile users, Internet of Things (IoT) nodes), where the UAVs are used as aerial base stations or relays. In [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], the UAVs are used as sensors and/or relays, where the goal is aerial search and coverage of a mission area. In some of these works, UAV-UAV connectivity is required, since the UAVs operate as a team, but UAV-to-ground connectivity is not necessarily included in the mission goals [26], [29], [30], [31], [32], [33], [34], [35]. Target monitoring generally appears as sensing of a point of interest or mobile targets; however, connectivity of targets is not considered [34], [36], [37], [38].

In this work, we consider all of these tasks in our SAR mission, where a team of UAVs is deployed to *search an area for multiple targets* and connect the detected targets to the ground control station (GCS) to *continuously monitor and service them*, while *maintaining the connectivity of the UAVs to GCS*. Since these mission tasks and the corresponding characteristics required in UAVs vary, we define two types of UAVs: search UAVs with data collection sensors and wireless interfaces and relay UAVs with wireless interfaces. While the search nodes are tasked with sensing the disaster area and detecting and monitoring the targets, relay nodes are tasked with connecting the UAVs and the targets to the GCS. We propose to treat the multi-target SAR mission in three phases, which are marked with 2 events: first target detection and last target detection. We optimize the paths of search and relay drones sequentially. In phase 1, search and relay nodes fly pre-defined paths. The search node paths are optimized such that the total traveled distance is minimized and the generated search paths are fed into our dynamic relay positioning algorithm, where the relay node paths are optimized such that the search nodes are connected to GCS at all times using a minimum number of relays [39]. Phase 2 is between the first and last target detection. We define detection as a target being in the sensing coverage of at least one search UAV. We propose 3 approaches as a response to each target detection: (i) Detection Priority Path Planning (DPPP), where search and relay drones fly their pre-planned paths till the last target detection and

Manuscript received 15 July 2023; revised 15 December 2023; accepted 24 January 2024. Date of publication 13 February 2024; date of current version 16 July 2024. This work was supported by the Scientific and Technological Research Council of Turkey (TUBITAK) ARDEB 1001 under Grant 121E408. The review of this article was coordinated by Prof. Shahid Mumtaz. (Corresponding author: Evşen Yanmaz.)

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Digital Object Identifier 10.1109/TVT.2024.3363840

inform GCS of each detection; (ii) Extended Detection Priority Path Planning (EDPPP), where search and relay nodes behave as in the DPPP, but GCS injects a new search node and sends it to the detected targets to improve target monitoring; (iii) Detection Adaptive Path Planning (DAPP) re-plans UAV paths at each target detection such that each target is monitored continuously and is connected to the GCS upon its detection. Phase 3 relates to the formation of a UAV network upon last target detection such that each target is covered by a search node and connected to the GCS with a minimum number of relay nodes. The first 2 phases plan the paths in a centralized manner. In the final phase, the UAVs within the same network clique plan their paths in a distributed manner, assigning the best node to open tasks. Each of the proposed responses to target detection has its advantages and disadvantages in terms of mission phase times, UAV network connectivity and target connectivity to GCS. While DPPP and EDPPP result in the lowest detection times, DAPP makes up for the lost time in detection time by faster re-organization to form the relay network in the final phase. During the mission, DPPP leads to the worst target connectivity since the priority is given to finding all targets. DAPP, on the other hand, allows continuous monitoring and very high connectivity. Furthermore, we investigate the performance difference of the proposed schemes with multi-objective optimization-based path planners using single type search UAVs. The aim is to jointly optimize the area coverage time and percentage connectivity of the UAVs to GCS. We propose two formulations and solution methods for this optimization problem: a) weighted-sum method and b) a non-dominated sorting genetic algorithm 2 (NSGA2) solution [40]. We first test these path planners in their basic forms, assuming all nodes are search nodes. Then, we incorporate the jointly optimized search path plans into our dynamic path planner workflow and use the joint paths as an input to the relay positioning algorithm for a fair comparison. Our results show that the basic paths without a relay lead to shorter traveled distances but perform significantly worse in terms of connectivity, especially when number of search drones is low. When the weighted-sum and NSGA2 paths are combined with the relays, percentage and target connectivity of the UAVs to the GCS become comparable to DPPP; however, the mission times may be up to 40% longer than DAPP.

Our contributions can be summarized as follows:

- We propose multi-UAV path planning algorithms for multiple target detection, monitoring, and connectivity in a SAR mission with different levels of adaptability while maintaining connectivity of the UAVs to GCS.
- We propose to use two types of drones: search drones for target detection and monitoring and relay drones for UAV-UAV, UAV-GCS, and target-GCS connectivity. We provide a workflow for our dynamic path planner, where the search and relay drone paths are optimized sequentially. With such an implementation, initial search node paths can be freely chosen and the proposed path planners can be used for different applications.
- As a benchmark, we also model the path planning problem as a multi-objective optimization problem and solve it by weighted-sum and NSGA2 solvers. We extend this solution by integrating it into our dynamic path planner workflow for a fair comparison in terms of used resources.

The remainder of the paper is organized as follows. Related work is summarized in Section II. In Section III, we provide the system assumptions. The UAV positioning algorithms are presented in Section IV. Results are given in Section V and the paper is concluded in Section VI.

II. RELATED WORK

Multi-UAV systems can be deployed for different missions and the UAVs can take on different roles based on the required tasks. In this work, the mission tasks consist of target search, UAV-UAV connectivity, UAV-GCS connectivity and multi-target connectivity to GCS. To maintain the connectivity of multiple static targets as well as highly mobile UAVs, we propose to use two types of drones. The inclusion of dedicated relay nodes with no sensing capability enables connectivity maintenance among network entities with different characteristics (e.g., static ground nodes, mobile UAVs).

A. Ground Target Connectivity-UAVs as Aerial Base Stations or Relay Nodes

We first present works where the UAVs operate as relays or base stations such that ground nodes are connected to the backbone or to each other. The ground nodes can be mobile users, sensors, IoT devices, etc. The UAV positions or trajectories are optimized such that network-related criteria are met. Possible relay positioning criteria include maximizing network coverage area and lifetime [2], minimizing the number of drones and energy consumption of ground users [3], [6], maximizing energy efficiency [4], [5], data transmission and coverage fairness [7], minimizing the number of UAVs [8], [9], maximizing network throughput [4], [41], maximizing minimum data collection speed [42], minimizing the number of UAVs and data rate dissatisfaction [43], optimizing target coverage ratio and charging operation scheduling [44], optimizing data service and flight time [45], maximizing minimum rate [46], maximizing connectivity [11]. In these works, the mission is focused on network coverage, where the main goal is connecting ground targets to the backbone, where all UAVs are aerial base stations or relays. UAV-UAV connectivity or searching and detecting multiple-targets are not considered.

B. Search and Coverage-UAVs as Search and/or Relay Nodes

In missions such as surveillance, search and rescue, and environmental monitoring, the objectives that UAVs need to accomplish can be diverse. Connectivity has been incorporated into the planners as a constraint, as an objective or has been assumed to be all-time available. In these works, the paths of the UAVs are jointly optimized. The UAVs can take on mission and/or relay roles, where the mission can be coverage, search, target detection, target tracking, etc. For the coverage and search applications, the main objectives include minimizing energy [12], [13], [14], minimizing travel time or path length [16], [17], [18], [19], minimizing search time [20], [21], maximizing target detection [20], [22], [23], maximizing information gain (or minimizing uncertainty) [21], [24], [25], maximizing area coverage [26], [27], [28], etc. These works mostly correspond to target search task considered in our work.

Coverage and search objectives are optimized under connectivity considerations in [26], [27], [29], [30], [31], [32], [33], [34], [35], [47], [48], where area search and UAV connectivity tasks are optimized jointly for mostly single type UAVs. Different types called relay, search, articulate, jammer, and collector are specified in [6], [27], for each UAV as a function of its distance from the GCS and other UAVs. In these works, mainly connectivity between UAVs is considered such that team or swarm behavior can be maintained. Connectivity to GCS is considered in [27], [30], [47], [48], [49]. In our work, we not only maintain UAV-UAV and UAV-GCS connectivity, but

TABLE I
SYSTEM PARAMETERS

System Parameters	Definition
N_s, N_i, N_r	Number of search drones, maximum injected and relay drones
N_{tar}	Number of targets
A, A_{uncov}	Total and uncovered mission area
r_s, r_r	Transmission and sensing ranges
V_s, V_r	Maximum search and relay drone speed
t_s	Time step duration
F	Maximum flight time
P_{GCS}	GCS position

also aim target monitoring and target connectivity to GCS. Target monitoring appears in works, where UAVs cooperate in tracking mobile ground nodes [34], [36], [37], [38]. In these works, however, the main goal is joint sensing of the target and connectivity to a ground control is not in focus.

C. UAV Path Planning as an Optimization Problem

The planning problem is generally modeled as a single or multi-objective optimization problem. Classic solvers such as integer linear programming [20], [45], weighted sum [2], [3], [12], [13], [16], [21], [22], [24], [25] and evolutionary and nature-inspired methods such as particle swarm optimization [15], ant colony optimization, genetic algorithm [14], [17], [26], [27], [29], [30], [32], NSGA2 or 3 [19], [28], [43] have been used for the path planning problem. Recently, machine learning and deep learning-based approaches have also been proposed for multi-UAV applications. Learning-based approaches strive to maintain connectivity by training UAVs to be connected throughout a collaborative mission like dynamic target tracking. [50] uses a combination of a neural network and genetic algorithm for a swarm of UAVs, jointly maximizing the search area coverage and connectivity maintenance. Unlike conventional methods, deep reinforcement learning algorithms try to teach each UAV to connect with other UAVs in a fleet of UAVs based on reward and punishment policies during a mission [6], [7], [10], [51], [52], [53], [54]. Recently, we have proposed a reinforcement-learning based path planner such that the UAV-GCS connectivity is maintained during a target search mission for single-type UAVs and illustrated the benefits of online decision-making in comparison to pre-planned multi-objective optimization based methods [55]. While learning-based methods have many merits, the multi-target detection and connectivity mission under consideration is complex and there is still not enough mission data to train the multi-UAV systems such that the mission can be reliably completed. Therefore, transitional schemes such as in this paper that incorporate both pre-planned planners and event-driven decision-making are expected to provide valuable insight without the need for complex training.

III. SYSTEM MODEL AND ASSUMPTIONS

A. Mission Environment and the UAVs

The system parameters are given in Table I. The area of interest is denoted by A and is assumed to be square and modeled as a grid with equally-sized, disjoint cells of size $r_s \times r_s$, where r_s is the ground sensing coverage area of the UAVs. We consider multi-rotor UAVs that are capable of waypoint flight and hovering. Search UAVs are equipped with various sensors,

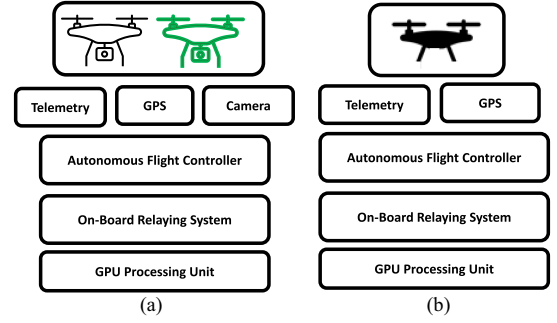


Fig. 1. Interfaces of the drones: (a) Search (white)-injected(green) and (b) Relay.

e.g., GPS to know their positions and a camera to observe the environment. All UAVs have a wireless communication interface to exchange information with other UAVs, and the GCS. Fig. 1 shows the interfaces of drones. The search and injected drones (Green) have similar interfaces. The maximum flight times of the UAVs are limited to F . A fixed number, N_s , of search nodes start the mission. Up to N_r relay and up to N_i search nodes can be injected into the system at different phases of the mission. N_{tar} targets are assumed to be randomly distributed in the mission area. The targets can be any point of interest (See Fig. 2), e.g., victim groups, damaged buildings, rescue personnel, etc. In this work, we focus on establishing connectivity among targets using a minimum number of nodes and shortest path routing.

During the mission, search UAVs continuously sense the area in their coverage and exchange information with the UAVs within its clique, i.e., with any search or relay UAV that can be reached over a single or multiple hops. The exchanged information can include detected target status, whether the GCS knows of target locations, target monitoring status, position information, mission updates, etc. Perfect sensing is assumed, i.e., when a cell is visited, it is sensed without uncertainty and if there is a target in that cell, it is detected with probability 1. We studied path planning with sensor uncertainties in [31]; however, it is out of the scope of this paper. The relay UAVs are not equipped with sensors, so they only exchange information as they meet other UAVs. We assume that the UAVs are equipped with the antenna setup given in [56] that has been shown to lead to free-space path loss in the absence of obstacles in all directions regardless of orientation and height differences. Thus, we assume a disc model for the communication links.

B. UAV Movement During the Mission

The multi-target search and connectivity mission consists of three phases: (i) pre-detection, (ii) post-detection (until the last target is found), and (iii) final configuration (after all targets are found). The search UAVs can move at most at speeds V_s and are not allowed to revisit an already covered cell during the pre-detection phase. Once targets are detected, some search nodes may hover over the targets and monitor them depending on the path planner used. The relay nodes can fly at speeds up to V_r and can hover or move to already visited locations to assist connectivity at any point during the mission.

The search nodes fly pre-planned or re-planned paths, given in Section III-C, depending on the mission phase. Relay nodes compute their positions at each time step, t_s , based on the positions of the search nodes and the targets. The relay nodes

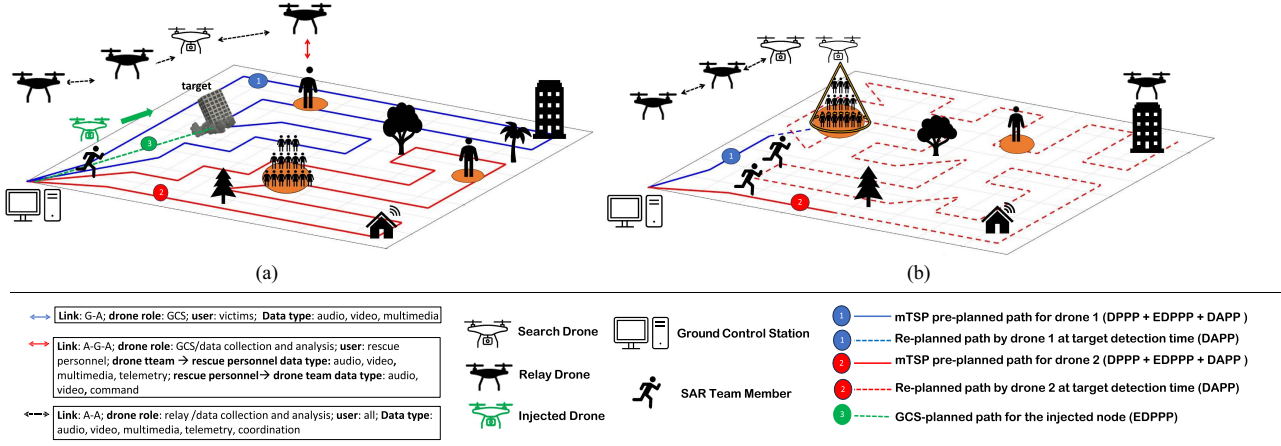


Fig. 2. Multi-target search and rescue mission. The blue and red solid lines indicate the pre-planned search paths for 2 drones. (a) In EDPPP, search nodes fly pre-planned paths until final target detection. A new node is injected (green drone) and sent to the detected target location. (b) In DAPP, a search node that finds a target hovers at the target, and the paths of the remaining drones are re-planned (indicated by the dashed lines).

are assumed to fly to their next position on a straight line at a fixed height, which can be adjusted for safety purposes.

C. Preliminaries

1) *Basic Search Path Planning:* Since we assume perfect sensing, the search path problem can be treated as a coverage problem [57]. Therefore, we model the search paths in the pre-detection phase as an optimization problem, where the total traveled distance is minimized under certain constraints. The optimization problem is formulated as follows:

$$\text{Minimize } \sum_{m=1}^{N_s} \sum_{i=1}^P \sum_{j=1}^P \sum_{k=1}^K X_{i,j,k}^m D_{i,j} \quad (1)$$

$$\text{s.t. } \sum_{j=1}^P X_{i,j,k}^m \in (0, 1), \forall i \in P, \forall m \in N_s, \forall k \in K, \quad (2)$$

$$\sum_{m=1}^{N_s} \sum_{i=1}^P \sum_{j=1}^P \sum_{k=1}^K X_{i,j,k}^m = \|P\|, \quad (3)$$

$$X_{i,0,k}^m = 1, \forall m \in N_s, \exists i \in P, \quad (4)$$

$$X_{0,i,k}^m = 1, \forall m \in N_s, \exists i \in P, \quad (5)$$

$$\text{Minimum tour length} = d_m \quad (6)$$

$$\text{Search drone speed} \leq V_s \quad (7)$$

$$\frac{1}{V_s} \max_{\forall m \in N_s} \sum_{k=1}^K X_{i,j,k}^m D_{i,j} \leq \mathbf{F} \quad (8)$$

$$\text{Number of drones} = N_s, \quad (9)$$

$$\text{Transmission range} \leq r_c, \quad (10)$$

$$\text{Sensing range} \leq r_s, \quad (11)$$

$$\text{Area of interest} = A \quad (12)$$

where $X_{i,j,k}^m$ is 1 if a drone m moves from cell i to j at time step k , otherwise it is zero. $D_{i,j}$ is the distance between cells i and j .

P and K represent the drone path sequences in terms of consecutively visited cell numbers and their lengths, respectively. The constraints in (2)–(4) limit the paths such that each cell is visited exactly once; the paths start and end at the GCS (i.e., node 0); each search drone is used and the path lengths of the drones are balanced, respectively. The constraints in (5)–(11) are related to the system and UAV assumptions explained in Sections III-A and III-B.

We treat this search path problem as a multiple traveling salesman problem (mTSP) and use a genetic algorithm-based solution. The mTSP is an extension of the classical Traveling Salesman Problem (TSP) [58]. It includes multiple salesmen that depart from the same city with the aim of visiting a set of cities with minimum total traveled distance. Fig. 2 shows the two initial mTSP search drone paths marked with red and blue solid lines that collectively cover the entire mission area.

2) *Basic Relay Path Planning:* Recently, we have proposed a pre-planned relay positioning algorithm (RPA) [39]. Pre-planned RPA takes positions of the mission UAVs as input and positions relay UAVs such that mission UAVs (e.g., search UAVs) are connected to the GCS throughout the mission with a minimum number of relays. At a given time step k , a network graph G_k can be formed with positions of the GCS, the search nodes, and relay nodes as its vertices. An edge exists between any node i and j if the Euclidean distance D_{ij} between them is less than r_c . The adjacency matrix A_k can be computed as:

$$A_k(i, j) = \begin{cases} 1, & D_{ij} < r_c, \\ 0, & \text{otherwise} \end{cases} \quad i, j = 1, \dots, N_s + N_r + 1 \quad (12)$$

Connectivity of the drones to GCS, i.e., node 1, can be determined using powers of A_k , Fiedler value, or depth-first search. Let's set $B_k = A_k^{N_s + N_r + 1}$. The first row of B_k can be used to determine connectivity to GCS as:

$$c_k(j) = \begin{cases} 1, & B_k(1, j) > 0, j = 1, \dots, N_s + N_r + 1 \\ 0, & \text{otherwise} \end{cases} \quad (13)$$

The positioning problem is modeled as the following optimization problem:

$$\text{Minimize Number of relay nodes} \quad (14)$$

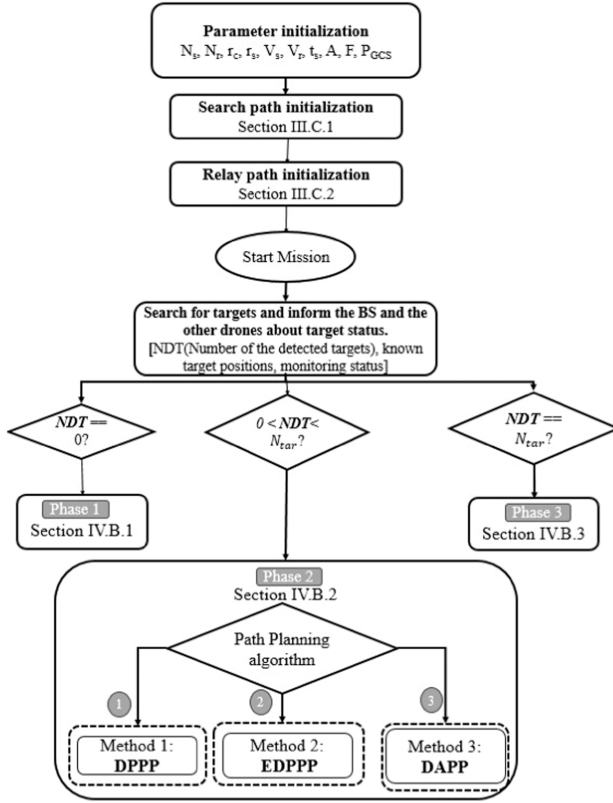


Fig. 3. Workflow of multi-target search and connect path planner.

$$\text{s.t. } \sum_{j=2}^{N_s+1} c_k = N_s, \forall k \quad (15)$$

$$\text{Relay drone speed} \leq V_r \quad (16)$$

$$\frac{1}{V_r} \max_{m \in N_r} \sum_{k=1}^K X_{i,j,k}^m D_{i,j} \leq F, \forall m \in N_r \quad (17)$$

$$\text{Transmission range} \leq r_c \quad (18)$$

where (15) enforces all mission nodes to be connected to GCS. The other constraints limit the speed, flight time, and transmission ranges. At a given time, the problem of connecting disjoint parts of the network can be considered as Steiner Tree Problem (STP). However, in the mobile scenario, the problem is not only finding the positions but also assigning feasible relays to the computed positions. The details and analysis of the pre-planned RPA can be found in [39]. In this paper, RPA will be used for the pre-planned relay UAV paths.

IV. UAV POSITIONING AND ASSIGNMENT ALGORITHM FOR MULTI-TARGET SEARCH AND CONNECTIVITY

A. Overview

The workflow of the proposed multi-target search and connectivity mission planner is shown in Fig. 3. The mission is designed in 3 phases. Until the first target detection, the search and relay drones fly pre-computed paths. From the first to last target detection, the detection event can cause different reactions; i.e., the proposed methods have different adaptability levels consisting

of pre-planned paths as well as event-driven paths. The last phase is setting up the drone network at a final configuration to monitor all targets continuously. Drones exchange information with the other nodes in its clique at each time step. The exchanged information includes position, target detection/monitoring status (i.e., which targets are detected and/or are being monitored), and GCS status (i.e., whether the GCS is informed of target status).

Search and relay nodes can be at one of the following states, which determine the actions that can be taken by the drone:

- Search nodes:
 - Searching: The drone has not found or been informed of any targets and not all targets are found.
 - Detecting: The drone has detected at least one target.
 - Monitoring: The drone is hovering over a target and collecting/delivering information.
 - Relaying: The drone is hovering at a relay position connecting targets to the GCS.
 - Unassigned: There is no open task, and the drone is flying home to GCS.
- Relay nodes:
 - Networking: The drone is connecting search and/or targets before all targets are found.
 - Hovering: The drone is hovering at relay positions connecting targets to the GCS.
 - Unassigned: There is no open task, and drone is flying home to GCS.

B. Drone Actions in Different Mission Phases

The UAV positions are computed for each time step according to their states. In the following, we will describe the potential actions depending on the drone states and target detection status at the end of each time step.

1) *Phase 1- From Take-Off Till First Target Detection:* The mission starts with pre-planned search and relay drone paths given in Section III-C (denoted as search and relay path initialization in Fig. 3). The search nodes are in searching and the relay nodes are in networking states. Algorithm 1 describes the path actions taken by the drones in the time interval from the mission beginning till the first target is found.

2) *Phase 2- From First Target Detection Till Last Target Detection:* Phase 2 of the mission is considered event-driven, where the event is the detection of targets. Between target detections, the search nodes are in searching and relay nodes are in networking states and follow their pre- or re-planned paths until a target is detected, exchanging information as they meet other drones as in lines 3–4 of Algorithm 1.

Once at least one target is found, the search drones are in detecting states. While in this state, they can take different actions and go into searching or monitoring states depending on the chosen path planning algorithm. In this paper, we propose path planners with different priorities and different reactions to target detection events. Each proposed algorithm addresses different requirements and can be applied to different UAV missions and targets.

- *Detection Priority Path Planning:* This basic method, proposed as a benchmark, prioritizes the searching task, where search and relay nodes fly pre-defined paths until the last target is found (see the blue and red solid paths in Fig. 2(a)). This method follows Algorithm 1 until the last target is found. The advantage of this path plan is

Algorithm 1: Pre-detection Search and Relay Path Procedure.

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- 1: **Procedure** PreSearchConnect $A, N_s, N_r, r_c, r_s, V_s, V_r, t_s, \mathbf{P}_{GCS}$
 - 2: Compute paths of search nodes using mTSP (Section III-C1)
 - 3: Compute paths of relay nodes using RPA (Section III-C2)
 - 4: At each time step, at each search and relay node repeat:
 - 4.1: Broadcast current position.
 - 4.2: Form a local network graph based on known drone positions and compute the node clique.
 - 4.3: Exchange GCS and target detection/monitoring status information with all nodes in the clique and update status information if necessary.
 - 5: At each time step, at each search node repeat:
 - 5.1: Sense the area within sensing coverage and check if a target is present.
 - 5.2: If a target is present, update the target detection status and move the detecting node and the nodes in its clique to post-detection phase (Section IV-B2) or final configuration phase (Section IV-B3); if not keep flying the current path.
 - 6: **end procedure**
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that no update to the mission plan is necessary until all targets are found; therefore, even if there are times the search drones get disconnected from the GCS they can still continue their mission. Since detection is prioritized, the search times are also minimized. However, the targets may be left unoccupied for a long time. Therefore, *this method is suitable for non-time critical environmental monitoring missions but might not be suitable for applications that require continuous monitoring or target connectivity.*

- **Extended Detection Priority Path Planning:** In this method, search and relay nodes still fly their pre-planned paths, but in order to obtain target information faster, GCS sends additional search nodes, that we call injected nodes, to monitor the detected targets. The injected node flies a straight line to the target position (e.g., in Fig. 2(a) while drones 1 and 2 fly their pre-planned paths shown by red and blue solid lines, the green drone is injected into the air when drone 1 finds a target). If, during its flight, other targets get detected, and the injected node becomes aware of it, it can be reassigned to another target position, depending on the total cost in terms of traveled distance. If the node arrives at its assigned target, it switches to the monitoring state and hovers over it, continuously sensing it. The original search and relay nodes follow lines 3 and 4 of Algorithm 1 until the last target is found. The actions taken at GCS and drones from the time of each detection are given in Algorithm 2. The main advantage of this method in comparison to DPPP is that the duration a target is unmonitored is significantly reduced. The disadvantage is more drones are in the air.
- **Detection Adaptive Path Planning:** In this method, every time a new target is found, the paths of relay and search drones are updated such that the detected targets are continuously connected to GCS as soon as they are detected, as shown in Fig. 2(b). The procedure at the detection time is given in Algorithm 3. Every time a target gets detected,

Algorithm 2: Extended Detection Priority Path Planning-EDPPP.

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- Input:** known target location(s), N_i, V_s
- Output:** paths from time of new target detection
- 1: From the time of detection till GCS is informed at each time step, at each search and relay node repeat lines 3 and 4 of Algorithm 1, respectively.
 - 2: At the time of detection, the detecting search node and its clique update the target detection status, GCS knowledge status and keep flying pre-defined paths.
 - 3: At the time GCS knows that at least one target is detected:
 - 3.1: GCS computes the shortest flight path from GCS to detected target location
 - 3.2: GCS injects a new search node in the air and assigns it to the newly detected target location
 - 3.3: Injected node flies the assigned path to the target. At each time step repeats step 5 of Algorithm 1. If it detects another target during its flight, it updates its target detection status and informs the nodes in its clique, starting the next event.
-

the detecting search node switches to monitoring state and stays at the target location (drone 1 with solid blue path in Fig. 2(b)). The other search nodes are still in the searching state, and their plan needs to be updated. At the time of detection, GCS runs an open mTSP (line 3.1 of Algorithm 3) over not yet covered cells, A_{uncov} . Therefore, the search nodes that are not covering a target start at their current locations, and the remaining uncovered cells are covered such that the total traveled distance is minimized without the need to go back to the initial locations (e.g., the path of drone 2 is updated to the red dashed line, when drone 1 detects a target in Fig. 2(b), while drone 1 hovers at the target position). Since, at this time, fewer search nodes will continue the searching (and in an effort to have a fair comparison with EDPPP), GCS also runs open mTSP with an additional injected node located at GCS (line 3.2). This step determines whether a replacement drone is necessary. As we shall discuss, especially when the number of original search nodes is low, this step becomes crucial. Once the search node paths are generated, the relay node paths are computed (line 3.4). The main advantage of this method is that targets are not left unoccupied. The available resources are more efficiently utilized. The disadvantage is that fewer search and relay nodes are available at every target detection to finish the mission. This method is especially suitable for UAV missions with ground target (such as mobile users) connectivity requirements.

3) **Phase 3- From Last Target Detection Till Mission End:** When all targets are found, the goal is to connect all targets to the GCS with a minimum number of nodes. At each target, a search node is present to sense and monitor. In this stage, the drones update their paths in a distributed manner. The drones that are aware of all target positions compute the final configuration using Algorithm 4. This algorithm treats the connectivity of the network of targets and GCS as the Steiner Tree Problem and uses the approximation in [59] as the solution. To this end, the Euclidean Minimum Spanning tree over target and GCS positions is computed (lines 2-3). Each long link is divided by r_c

Algorithm 3: Detection Adaptive Path Planning-DAPP.

Input: known target location(s), current search and relay node positions, A_{uncov} , number of unassigned search nodes, N_r , N_i , r_c , r_s , V_s , V_r

Output: paths from the time of new target detection

- 1: From the time of detection till GCS is informed at each time step, at each search and relay node repeat lines 3 and 4 of Algorithm 1, respectively.
- 2: At the time of detection, at detecting node and its clique update the target detection status and GCS knowledge status and start hovering until re-planning is complete.
- 3: At the time GCS knows that at least one target is detected:
 - 3.1: GCS computes paths of search nodes still in searching state ($path_{searchonly}$) using mTSP given in Section III-C1 without the return to start point constraints, where number of drones is number of unassigned search drones, and area of interest is A_{uncov} . GCS computes the mission duration for the computed paths ($time_{searchonly}$).
 - 3.2: GCS computes paths of search nodes still in searching state and an additional injected node (which is at GCS location, $path_{withinjected}$) using mTSP as above. GCS computes the mission duration for the computed paths ($time_{withinjected}$).
 - 3.3: GCS compares $time_{searchonly}$ and $time_{withinjected}$. If $time_{searchonly}$ is smaller, the re-planned search paths are $path_{searchonly}$; otherwise, GCS injects a new node in the air and returns the $path_{withinjected}$.
 - 3.4: GCS computes paths of relay nodes using RPA (Section III-C2) with re-planned search paths and the hovering search node locations as the set of nodes to be connected and the current relay positions as input and returns $path_{relays}$.
 - 3.5: GCS informs the search, relay and injected (if any) nodes of their new plans.
 - 3.6: The drones resume their mission.

Algorithm 4: Final Configuration Computation.

- 1: **function** finalConfig (target locations, P_{GCS} , r_c)
- 2: Form the Euclidean graph G with target and GCS locations as vertices and Euclidean distance d_{jk} between each vertex as the edge weights. Node 1 is assumed to be the GCS.
- 3: Compute Euclidean MST T for G , starting at node 1.
- 4: Find edges e in T with weights $w(e)$ larger than r_c and insert $\lceil w(e)/r_c \rceil - 1$ Steiner points into T .
- 5: Return the computed Steiner points.
- 6: **end function**

to compute the relay positions (line 4). The rest of the procedure in phase 3 is described in Algorithm 5. At each time step, each drone decides its next action together with the other nodes in its clique. Until all positions in the final configuration are occupied, the best node is assigned to open positions (Algorithm 5 line 12, 15). If a node is not assigned to a final configuration position, it starts flying back (line 10). If a node is assigned to a position, but on its flight toward the destination, it meets other nodes that might complete the assignment in a shorter time, it can be reassigned or become unassigned and fly to GCS.

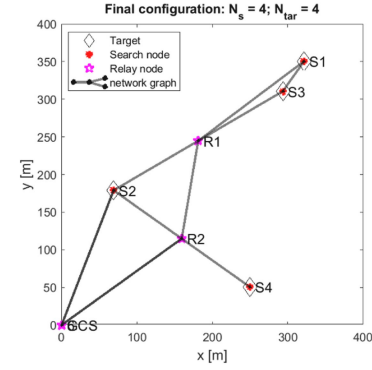


Fig. 4. Desired final configuration at the end of the mission. Grey lines show the network graph; $\diamond$ s are the 4 randomly generated target locations; S1-S4 and R1-R2 are the search and relay nodes, respectively.

MinCostTask takes as input the open positions and current node positions in a clique. The cost matrix is computed as the pairwise distances between positions (lines 4–5). A linear assignment method is used to match nodes to open positions such that the total cost is minimized (line 8). If the open position is a target position, only current positions of search nodes in a clique are considered in the matching. If the open position is a relay position, all nodes in the clique that are not already in a final configuration position are considered.

C. Illustration of Mission Phases

In this section, we illustrate node positions at different stages of the mission for the proposed methods. For this purpose, we assume 4 randomly distributed targets and 4 search nodes. The transmission range is assumed to be 200 m, and 3 relays are then sufficient to provide connectivity to the search nodes throughout the mission. Fig. 4 shows the final network. Each target (shown by $\diamond$) is covered by a search node (S_i), and 2 relays (R_i) are positioned such that the targets are connected to the GCS with a minimum number of relay nodes. Fig. 5(a) and (b) show the pre-planned and flown paths using DPPP and EDPPP methods. The search and relay nodes start the mission with their pre-planned paths computed using mTSP and RPA for all methods, and the missions end with the final configuration. In both methods, all 4 search nodes fly their pre-planned paths until the last target is found while exchanging information among themselves and the GCS with the help of R1, R2, and R3. With EDPPP, in comparison to DPPP, 2 injected nodes are also utilized. When S2 finds the first target, it informs the GCS, and GCS sends IN1 to the target location while S2 flies its path. When the second target is found by S4, GCS injects IN2. While IN2 is on its way to the target, the remaining targets are found, and the final configuration phase commences. S1-S4 and R1,R2 are assigned to targets and relay positions, respectively, using the MinCostTask algorithm. The unassigned R3 and IN2 fly back to GCS.

Fig. 5(c) and (d) show the flown search paths when DAPP is used with or without injected nodes. The line width is used to indicate the re-planned paths such that the line width is reduced each time a drone's path is re-planned (i.e., at each target detection). S2 finds the first target and it flies to the target location. When no injected nodes are used, paths of S2-S4

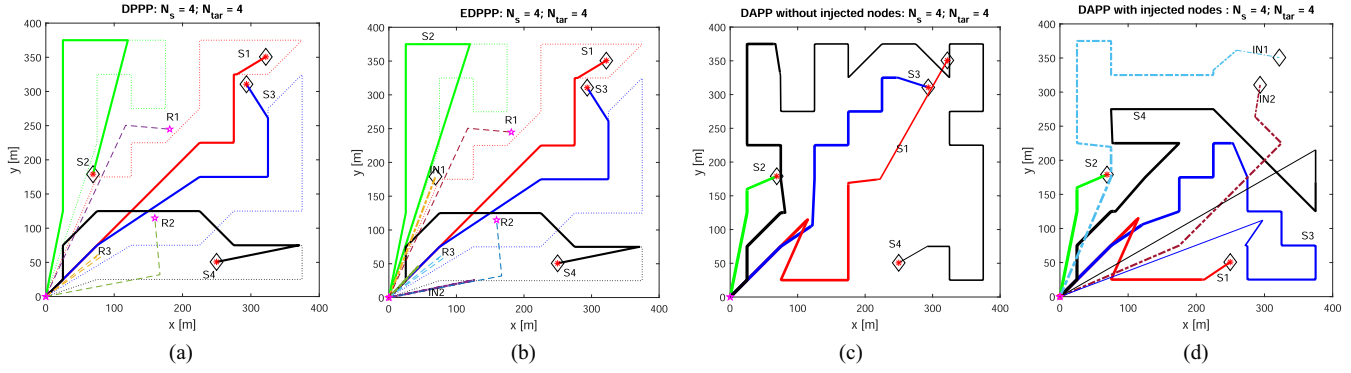


Fig. 5. Dotted lines show the pre-detection search paths computed using mTSP. The solid lines are the flown paths by the search nodes (S1–S4). The dashed lines are the flown paths by the relay nodes (R1–R3). Dash-dotted lines show the paths flown by injected nodes (IN1–IN2). In [c,d], the thickness of the lines decreases, indicating the re-planned paths (a) Mission with DPPP. (b) Mission with EDPPP. (c) Mission with DAPP without injected nodes. (d) Mission with DAPP with injected nodes.

and the connecting relays are re-planned using open mTSP (Fig. 5(c)). When injected nodes are used, re-planning includes IN1. Without injected nodes, S4 flies the longest path, covering the remaining parts of the hovering search nodes. When IN1 and IN2 are utilized, the path lengths are more balanced. The benefit of INs is more prominent when there are few search nodes left to cover a large area, as expected. The need to inject new nodes diminishes as the mission progresses.

V. RESULTS AND DISCUSSIONS

In this section, we analyze the performance of the proposed search and relay positioning approaches for multi-target detection and connectivity. The search region is 400 m × 400 m. The maximum search and relay node speeds are 2.5 m/s and 10 m/s, respectively. The sensing and transmission ranges are 50 m and 100 m, respectively. These parameters are chosen based on our experimental work [56], [60]. Typical drone specifications are used for speeds and transmission ranges, whereas r_s is chosen based on camera ground coverage with acceptable image quality. Considering a camera typical on commercial drones with a focal length of 28 mm and an image sensor of 1/2.3", the flight height can be computed as 200 meters based on the photogrammetry formulas [61] to acquire high-quality images and cover a cell region of 50 m × 50 m. N_r is limited to 20, whereas N_i is limited to $N_{tar} - 1$ for EDPPP and is limited to 1 for DAPP unless otherwise noted. The simulations are run in MATLAB for over 1000 randomly distributed target configurations.

A. Performance Metrics

We define the following performance metrics of interest related to mission phase times and connectivity:

- *Percentage connected time* is the ratio of time search drones are connected to GCS over the search time.
- *Detection time* is the time all targets are detected.
- *Mission time* is the time from take-off till the final configuration is reached.
- *Target connected time ratio* is the ratio of the time duration a target is connected to the GCS after its detection until the final configuration is reached.
- *Number of relay nodes* indicate the average and maximum number of relay drones in the air throughout the mission.

- *Number of injected nodes* is the number of additional search drones injected into the mission.
- *Distance traveled* is the (total or maximum) distance traveled by all drones.
- *Disconnected time* is the consecutive time duration a search node is disconnected from the GCS.

B. Benchmark Planners

The following algorithms are used to compute the initial path plans, where the first 3 approaches generate paths for search drones without relays:

- *mTSP*: The multi-UAV path planning problem is modelled as a single objective optimization problem, where the objective is to minimize the total travelled distance (see Section III-C1). The problem is solved using genetic algorithm.
- *Weighted sum (WS)*: The multi-UAV path planning problem is modelled as the weighted sum of two objectives, where the objectives are to minimize the total coverage time and maximize percentage connected time. We give equal weights to each objective and solve the optimization problem using genetic algorithm.
- *NSGA2*: The multi-UAV path planning problem is modelled as a multi-objective optimization problem with the same two objectives as WS. Both objectives are simultaneously optimized using NSGA2. Among the generated Pareto-optimal set of solutions, we choose the path plans with the best connectivity in our analysis.
- *mTSP+RPA, WS+RPA, and NSGA2 + RPA*: In addition to their basic forms without relay, we also analyze mTSP, WS and NSGA2 search paths integrated into our workflow given in Fig. 3 using method 1. To be precise, we first initialize search paths using mTSP, WS, or NSGA2 and for these initial paths we use pre-planned RPA to generate the relay paths. Note that mTSP+RPA following method 1 corresponds to DPPP.

Table II shows the worst-case algorithmic complexities of the proposed methods. The complexity of DPPP is determined by the number of generations (G), population size (P) the number of drones (N_s) and the number of cells ($m = A/r_s^2$). To compute connections between drones, we check each drone's position with respect to the other drones. Since we compute worst-case

Algorithm 5: Final Configuration Phase: From Last Target Detection to Mission End.

Input: Target, drone, and GCS positions, r_c, r_s, V_s, V_r, A
Output: Drone paths from last target detection until all nodes in final position

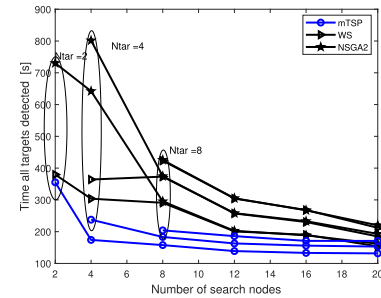
- 1: **while** All positions in final configuration not yet occupied, repeat on each Drone **do**
- 2: Information exchange
 - Broadcast current position.
 - Form a local network graph based on known drone positions and compute the Drone clique.
 - Exchange target detection/monitoring status and GCS status with all nodes in the clique and update status if necessary.
- 3: **if** Drone knows all targets **then**
- 4: **if** Drone knows GCS is informed of all targets **then**
- 5: **if** Drone is in final configuration **then**
- 6: Drone stays where it is.
- 7: **else**
- 8: **if** Drone knows all target positions are covered **then**
- 9: **if** Drone knows all relay positions are covered **then**
- 10: Drone is unassigned and it flies back to GCS.
- 11: **else**
- 12: Assign the nodes in the connected component of Drone to open relay positions using the function $\text{MinCostTask}()$. Generate the direct paths corresponding to the assignment.
- 13: **end if**
- 14: **if** Node is a search or injected node **then**
- 15: Assign the search or injected nodes in the connected component of Drone to open target positions using the function $\text{MinCostTask}()$. Generate the direct paths corresponding to the assignment.
- 16: **end if**
- 17: **end if**
- 18: **end if**
- 19: **else**
- 20: Drone flies toward GCS to inform it about all targets.
- 21: **end if**
- 22: Drone flies its current path.
- 23: **end if**
- 24: **end while**

Algorithm 6: Minimum Cost Assignment to Open Positions.

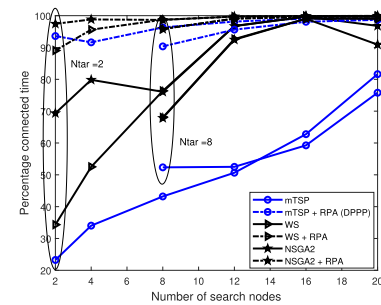
- 1: **function** MinCostTask (open positions, current positions)
- 2: **for** each current relay position j **do**
- 3: **for** each open position k **do**
- 4: Compute Euclidean distance d_{jk} between position j and position k
- 5: Set cost $\text{Cost}(j, k) = d_{jk}$
- 6: **end for**
- 7: **end for**
- 8: Match a current position to an open position such that total cost is minimized solving the linear assignment problem.
- 9: **end function**
- 10: Return matched location pairs.

TABLE II
COMPLEXITY OF THE PATH PLANNERS

Algorithm	Estimated Time Complexity
DPPP	$O(G \times P \times (N_s \times m + N_s^2))$
EDPPP	$O(N_{tar} \times G \times P \times (N_s \times m + N_s^2))$
DAPP	$O(N_{tar} \times G \times P \times (N_s \times m + N_s^2))$
NSGA2	$O(P \times \log(P) \times G \times (N_s \times m + N_s^2))$



(a)



(b)

scenarios, we consider that all drones are connected for this computation. We also determine the distance each drone flies during the mission by checking whether they visited the cities in their path or not. For EDPPP, path recalibrations occur for the number of targets (N_{tar}). Similar to EDPPP, DAPP also recalibrates paths for all targets at the worst-case. NSGA2 uses non-dominated searching which has a complexity of $P \times \log(P)$ [62].

Fig. 6 shows the target detection times and the percentage connected times of mTSP, WS and NSGA2 schemes for different N_s and N_{tar} values. Note that the detection times with or without RPA are the same, therefore the detection results for WS+RPA and NSGA2+RPA are omitted from Fig. 6(a) for

Fig. 6. Comparison of detection times and connectivity vs number of search nodes for benchmark search algorithms. (a) Time all targets detected vs N_s . (b) Percentage connectivity of search nodes vs N_s .

better readability. Since mTSP aims to minimize total traveled distance and correspondingly coverage time, its target detection time is the shortest. On the other hand, since NSGA2 and WS incorporate connectivity in the mission plan, their percentage connected times are significantly better than mTSP. Therefore, in an effort to keep the drones connected to GCS, the detection times are increased for WS and NSGA2. Since NSGA2 chooses the best-connected solution among the Pareto-optimal solution

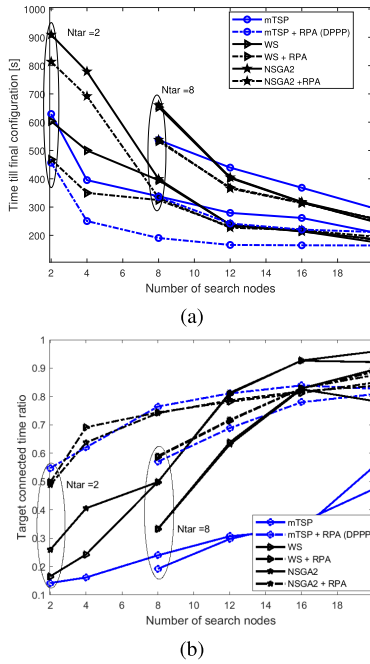


Fig. 7. Comparison of mission time and connectivity metrics vs number of search nodes for benchmark search algorithms. (a) Time final configuration reached vs N_s . (b) Target coverage time ratio vs N_s .

set, it leads to better-connected paths than WS but, conversely, longer detection times.

Fig. 7 shows the mission time and target connectivity performance of mTSP, WS, and NSGA2 schemes both in their native form and with RPA. The use of relay drones improves the performance of all schemes since the target detection information is readily available to GCS and the other drones and the time to reach the final configuration is reduced. Observe that DPPP (i.e., mTSP+RPA) leads to better mission times than WS and NSGA2. Therefore, the use of relays compensates for the time lost to inform the GCS and mission time is accordingly reduced below the other schemes.

For a fair comparison, in the rest of the paper, we analyze the RPA-amended versions of the WS and NSGA2 schemes.

C. Required Resources

We first determine the number of additional relays and injected nodes utilized during the search and connect mission. Fig. 8 shows the average and maximum used relay nodes and the average number of injected nodes. Observe that while there are more relay nodes available, a maximum of 11 relay nodes are used, and on average, the number of used relay nodes changes between 3 and 5 when the number of search nodes is low (see Fig. 8(a) and (b)). As the number of search nodes increases, the connectivity of the search node network also increases, and fewer relay nodes are required to keep them connected to the GCS. DPPP and EDPPP lead to similar performance as the search and relay node paths are the same until the last target is detected; DAPP, on the other hand, uses fewer relay nodes. For WS and NSGA2 with RPA, the average number of relays required is less. This is due to the fact that both search node path plans already take into account search node connectivity in the initial planning stage. Fig. 8(c) shows the average number

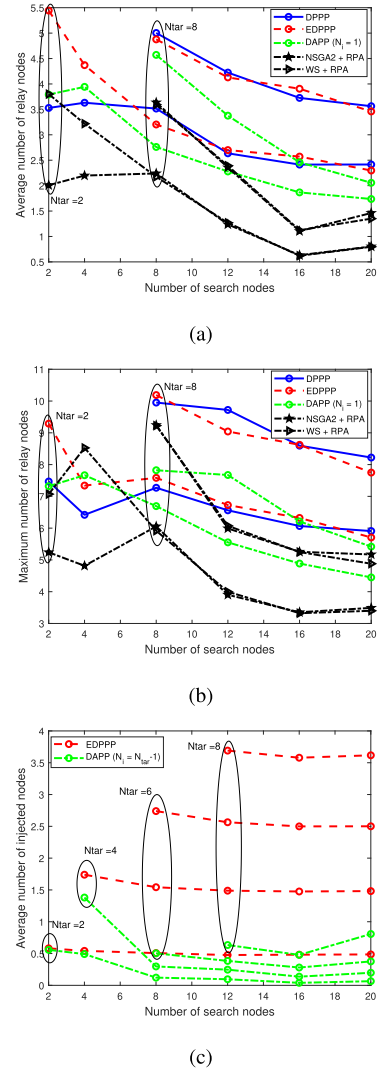


Fig. 8. Comparison of resources used vs number of search nodes. (a) Average number of relay nodes vs N_s . (b) Maximum number of relay nodes vs N_s . (c) Average number of injected search nodes vs N_s .

of injected nodes for EDPPP and DAPP when $N_i = N_{tar} - 1$. Observe that for EDPPP, around half of the available injected nodes are utilized, whereas for DAPP, on average 1 node is injected to assist the mission. Therefore, unless the targets are found early enough that the uncovered area is large, node injection is not required. In the following results, N_i will be limited to 1 for DAPP.

Fig. 9 shows the total and maximum traveled distance of all nodes deployed in the mission. Overall the total traveled distances are comparable to each other, whereas WS and NSGA2 have lower distances. One of the objectives that both methods optimize is the total coverage time, which is closely related to the traveled distances. On the other hand, maximum distance for NSGA2 for low N_s is higher than the other schemes. This is due to the fact that some search nodes have to cover larger areas while the other nodes assist connectivity. For higher number of targets DAPP has a higher maximum distance since as targets are found, the remaining paths of the detecting search nodes need to be covered by the others.

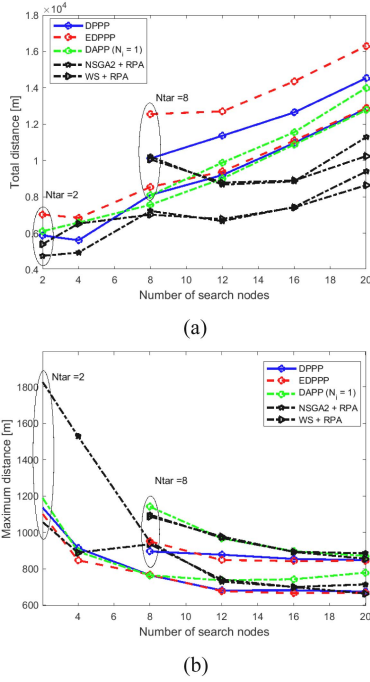


Fig. 9. Comparison of path lengths vs number of search nodes. (a) Total traveled distance vs N_s . (b) Maximum traveled distance vs N_s .

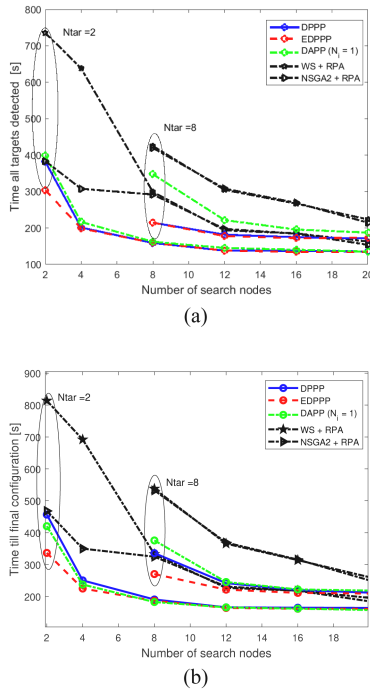


Fig. 10. Comparison of mission phase times vs number of search nodes. (a) Time all targets detected vs N_s . (b) Time final configuration reached vs N_s .

D. Mission Phase Times

Next, we present the duration of different mission phases. Fig. 10 compares the time till the detection of all targets and the time till the final configuration is reached for different numbers of search drones and targets. As expected, the detection times for EDPPP and DPPP are the same since the search node paths

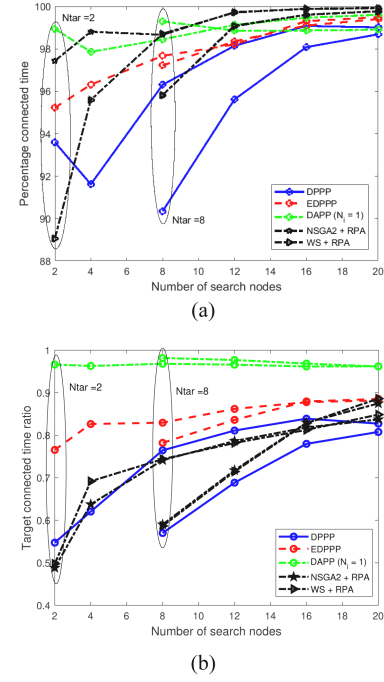


Fig. 11. Comparison of connectivity metrics vs number of search nodes. (a) Percentage connectivity of search nodes vs N_s . (b) Target coverage time vs N_s .

are pre-defined. When the number of targets and drones are the same, DAPP takes longer to detect all targets since fewer drones are available for search after every target detection. As the number of drones increases for a given N_{tar} , the performance of detection times of all schemes converge. WS and NSGA2 schemes lead to longer mission times, since the drones do not spread in the coverage area as mTSP in order to keep the drones connected to BS.

As we compare the total time from take-off until final configuration, where all targets are monitored by a search drone and connected to GCS with a minimal number of nodes, we can see that the extra time spent on search in DAPP is compensated almost immediately by the formation of a relay network. For instance, for DPPP, when $N_s = 2$ and $N_{tar} = 2$ the target detection time is around 350 s, and the time to form the relay network is around 100 s, making the total time 450 s, which is higher than that of DAPP. So, although all targets are found faster with DPPP, the information transmission (monitoring phase) does not commence immediately. EDPPP has the advantage of having the injected nodes already at the targets (even though possibly not connected to GCS), which reduces the relay formation time, leading to a better total time. The difference in total detection time affects the overall mission time for WS and NSGA2 schemes as expected.

E. Connectivity

In the following, we compare the performance of the proposed path planners in terms of drone and target connectivity. Fig. 11(a) shows the connectivity of the search nodes to the GCS from take-off till the final configuration is reached. The percentage connected time is above 85% for all proposed schemes. Since there are enough relay nodes, high connectivity is to be expected during the mission. Disconnections may happen in the last part of the mission when the nodes are organizing to move to their

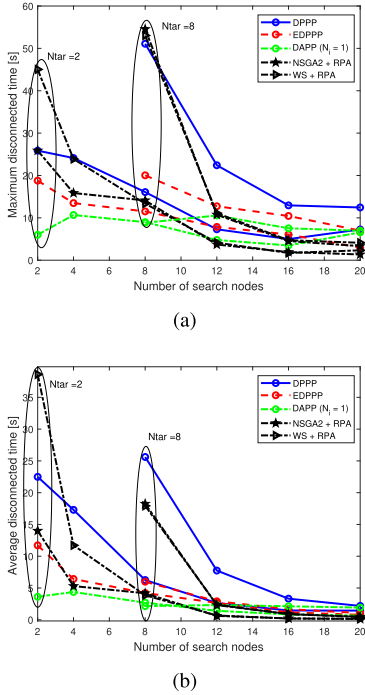


Fig. 12. Comparison of disconnected times vs number of search nodes. (a) Maximum disconnected time vs N_s . (b) Average disconnected time vs N_s .

final positions. DAPP's percentage connected time is the highest given that all but the last detecting search node is in place and connected to the GCS over a minimum number of relay nodes, and open node positions are minimal. For DPPP, very few search nodes are likely to be close to a target position when the last target is found. This is especially the case when N_s is small. During the reorganization, connectivity is not guaranteed, so the percentage connectivity is lower. As N_s increases, connectivity improves significantly due to a denser network. EDPPP lies between DPPP and DAPP, since injected nodes assist connectivity.

Fig. 11(b) shows the connectivity of the detected targets to the GCS from their detection time till the final configuration is reached. As expected, DAPP has almost 100% target connectivity (disconnection may happen due to final reorganization, as explained above). DPPP and EDPPP may have severe target connection disruptions, even leading to ratios less than 50% depending on N_s and N_{tar} . EDPPP shows better performance than DPPP, since the injected nodes monitor the targets and enable data delivery to GCS, even in intermittent connectivity scenarios. Therefore, depending on the application's tolerance to delays in updates on the target, choice of the path planning approach can change. WS and NSGA2 perform similar to DPPP.

We can further analyze the connectivity levels by measuring the consecutive time interval during which the search drones get disconnected from the GCS. Fig. 12 shows the average and maximum disconnected times of the search nodes for different scenarios. DAPP leads to connected search nodes throughout the mission, with disconnections of 5 s on average occurring in the last stage of the mission. DPPP has, on average higher disconnected times (~ 25 – 40 s for small N_s). As N_s increases, the disconnected times of all schemes converge due to higher spatial density and overall connectivity. The WS may also suffer from longer disconnections when N_s is small, whereas NSGA2 perform comparable to DPPP. However, with increasing N_s , all methods have shorter disconnected times.

VI. CONCLUSION

We have proposed and analyzed multi-UAV path planners for multi-target search and connectivity with different levels of adaptability to target detection. We have compared our proposed planners with joint optimization-based planners, where all UAVs are both search and relay UAVs and the paths are optimized in terms of total coverage time and percentage connectivity. Our results show that the less adaptive schemes, DPPP and EDPPP, detect all the targets faster; however, the connectivity of the UAVs and the targets to GCS is significantly better for more adaptive DAPP. The weighted-sum and NSGA2 methods, on the other hand, in their basic forms, trade off detection times to improve connectivity. When WS and NSGA2 are used in conjunction with relays, performance is improved; however, overall mission times and connectivity are still not as good as the proposed schemes. Therefore, while initial pre-planned paths are important, depending on the tasks to be achieved, the available resources might be utilized more efficiently if some tasks are handled by dedicated UAVs during different phases of the mission. Our future work will address time-limited target detection and target tracking. Therefore, we aim to extend the current work to a more online form utilizing classic and deep learning methods, where the path decisions are made on the fly.

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